

ATP Match Duration and Ace Performance Analysis

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Introduction

This project analyzes 2024 ATP tennis match data to study whether match duration can be predicted using player and match variables, and whether winner ace performance can be classified as high or low using player and match characteristics.

This analysis could be useful for TV broadcasters planning match coverage and for coaches studying opponent serving patterns.

Research questions

1. **Regression question:** Can match duration be predicted using player, match, and tournament variables?
2. **Classification question:** Can winner ace performance be classified as high or low based on player and match variables such as winner height, surface, match duration, ranking, first-serve performance, and handedness?

Key results

- Multiple Linear Regression explained about 31% of the variation in match duration.
- KNN regression performed best at approximately $K = 16$.
- Multiple Linear Regression had slightly lower test error than KNN and was easier to interpret.
- Logistic Regression and QDA both achieved roughly 71–73% cross-validation accuracy for classifying ace performance.

Data set description

The dataset used for this analysis is the *2024 ATP Match Results* database. It is maintained by *Jeff Sackmann*. It provides a record of every professional match played on the ATP Tour for the season.

Source

Jeff Sackmann's Tennis ATP GitHub.

URL:

https://github.com/JeffSackmann/tennis_atp

Collection:

The data is compiled from ATP match reports and tournament logs. It includes matches from all surfaces and various tour levels (Grand Slams, Masters, etc.) Additionally, the data includes quite a few in-game statistics.

Statistical Methods

Regression

To answer the regression question on match duration, I use two regression techniques: Multiple Linear Regression and K-Nearest Neighbors regression.

Multiple Linear Regression

The first regression technique used is Multiple Linear Regression. I chose multiple linear regression because it is a standard and highly interpretable method for predicting a quantitative response. In this project, it allows me to examine how match duration changes with predictors such as winner's aces and court surface, while also quantifying the direction and size of each predictor's effect. Multiple linear regression is especially useful because its coefficients can be interpreted directly, making the relationship between the predictors and match duration easy to explain.

For this technique, I use backward stepwise selection to identify which predictors remain in the final model. After running backward selection, the resulting model is summarized below.

Call:

```
lm(formula = minutes ~ w_ace + surface + winner_ht + loser_rank +  
    best_of + draw_size + winner_age, data = atp.mlr)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-119.332	-24.756	-4.042	22.396	146.483

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	173.382040	21.423510	8.093	8.62e-16	***
w_ace	2.279634	0.147763	15.428	< 2e-16	***
surfaceClay	12.406385	1.592813	7.789	9.47e-15	***
surfaceGrass	-6.954245	2.168840	-3.206	0.00136	**
winner_ht	-0.894040	0.109693	-8.150	5.43e-16	***
loser_rank	-0.012583	0.004183	-3.008	0.00265	**
best_of	24.299653	1.262627	19.245	< 2e-16	***
draw_size	0.035932	0.018582	1.934	0.05325	.
winner_age	0.405162	0.152525	2.656	0.00794	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 34.34 on 2769 degrees of freedom

Multiple R-squared: 0.3095, Adjusted R-squared: 0.3076

F-statistic: 155.2 on 8 and 2769 DF, p-value: < 2.2e-16

The model summary shows two main results. First, the adjusted R^2 and multiple R^2 indicate that the model explains about 31% of the variation in match duration. Second, the F-statistic shows that the model as a whole is statistically significant, with a p-value of $2.2e-16$, which is far below 0.05. The fitted model can be written as follows:

$$minutes_i = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \beta_6 X_6 + \beta_7 X_7 + \beta_8 X_8 + \epsilon_i$$

where X_1 is the number of aces served by the winner, X_2 is an indicator variable for clay surface, X_3 is an indicator variable for grass surface, X_4 is the winner's height, X_5 is the loser's rank, X_6 is the best-of match format, X_7 is the draw size, X_8 is the winner's age, and ϵ_i is the error term for observation i .

Before interpreting the model further, I check the regression assumptions.

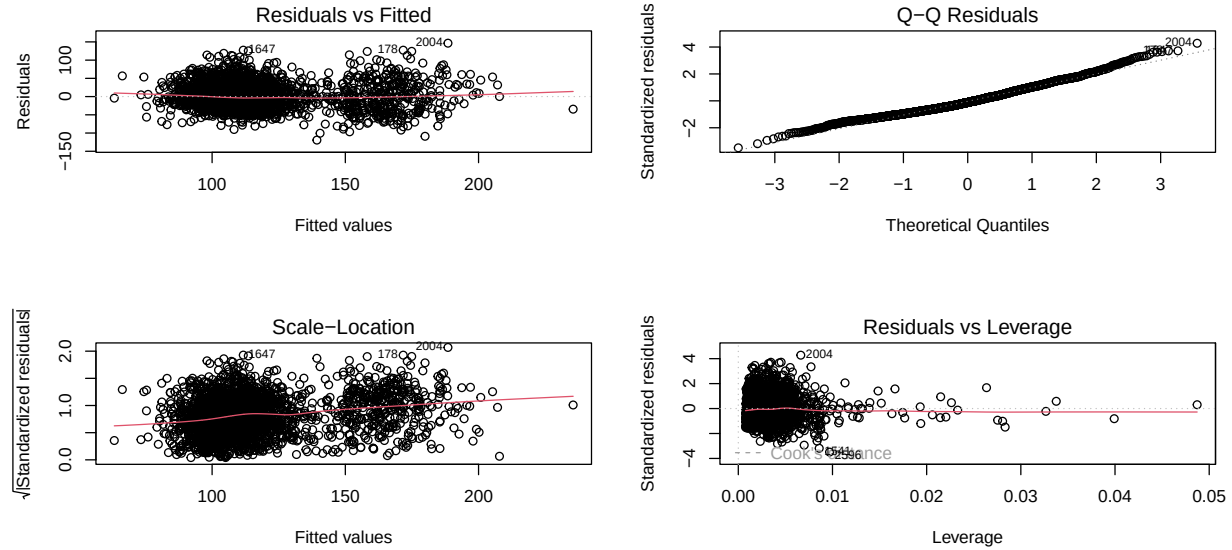


Figure 1: Diagnostic plots for the multiple linear regression model

Residuals vs Fitted

The residuals are centered around 0. However, there is some visible pattern across the fitted values. This suggests that linearity is approximately satisfied, while the data may be slightly heteroskedastic.

Normal Q-Q plot

The points are fairly close to the line in the middle of the Q-Q plot. However, there is deviation around the tails. Therefore, the data does not perfectly satisfy the normality assumption, but the residuals appear reasonably normal for this analysis.

Scale-Location

The red line is not perfectly flat, which supports the earlier observation that mild heteroskedasticity may be present.

Residuals vs Leverage

Most points have low leverage, with a few somewhat unusual observations. No observations appear to have extremely high leverage or influence.

Conclusion of Regression Assumptions

The plots indicate that the regression assumptions are reasonably met overall. The model is acceptable for this analysis, although the diagnostic plots suggest mild heteroskedasticity and some tail deviations from normality.

Before calculating the test error for MLR, I fit a KNN model and then compare the test errors for both regression models.

I also visualize the relationship between winner aces and predicted match duration. Since a multidimensional plot would be difficult to interpret, I hold the other predictors constant and vary only w_{ace} . I chose w_{ace} because it has one of the smallest p-values in the model and is interesting to examine visually.

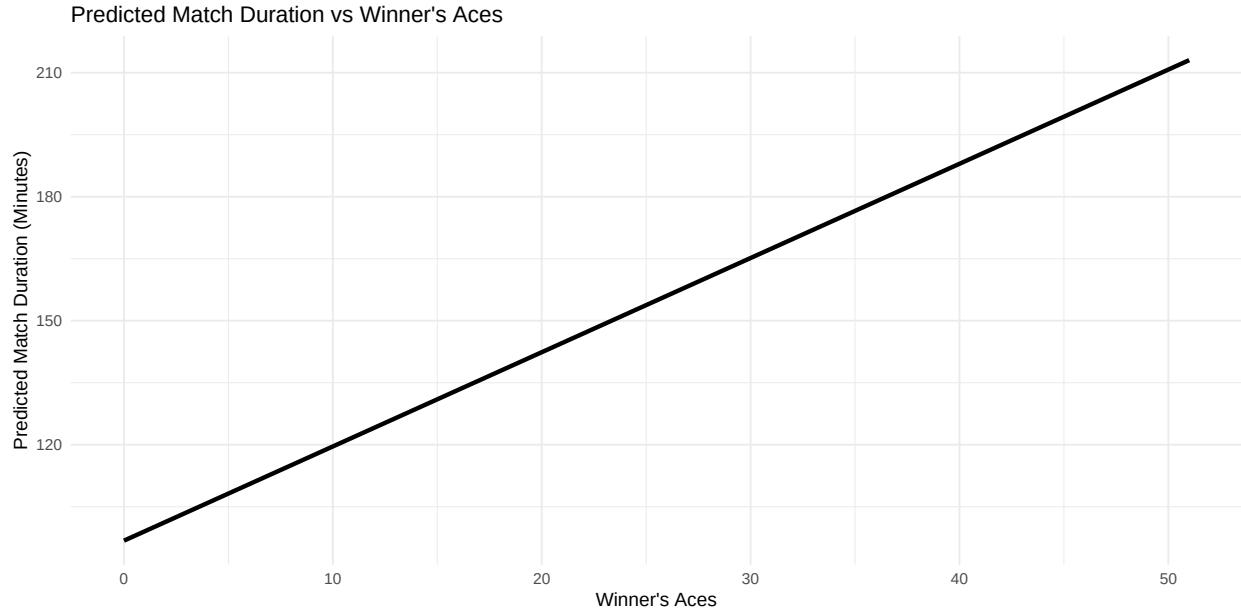


Figure 2: Predicted match duration by winner aces, holding other predictors constant

Holding other variables constant, predicted match duration increases as the winner's aces increase. This was initially counterintuitive. However, a possible explanation is that more winner aces often occur in longer matches, because longer matches give the winner more service opportunities.

K-Nearest Neighbors Regression

For the second regression technique, I fit a K-Nearest Neighbors (KNN) regression model. I used the same predictors as the MLR model because they are reasonable factors that could influence match duration, including player characteristics, match format, and tournament context. Using the same predictors also makes the comparison more fair.

The plot shows that model performance improves as K increases until roughly 15, after which performance begins to worsen slightly. The exact cross-validation results are shown below.

	k	RMSE	Rsquared	MAE	RMSESD	RsquaredSD	MAESD
1	1	48.80282	0.0896870	38.40422	2.175817	0.03725121	1.6052425
2	2	42.22780	0.1363220	33.24868	2.289540	0.04786723	1.5612693
3	3	39.72465	0.1667118	31.42552	2.109011	0.05063827	1.4497491
4	4	38.30886	0.1922412	30.31252	1.731969	0.04942562	1.1729747
5	5	37.48227	0.2099922	29.77739	1.410706	0.05390348	0.9834797
6	6	37.05436	0.2198810	29.36791	1.432979	0.06172529	1.0063305
7	7	36.54593	0.2325976	28.96216	1.374607	0.06419415	1.0327035
8	8	36.37478	0.2360514	28.78313	1.401753	0.06482350	1.0888890
9	9	36.03850	0.2468947	28.64872	1.368510	0.06449622	1.0255221
10	10	35.91433	0.2499783	28.56249	1.301682	0.06278393	0.9878387
11	11	35.86987	0.2511246	28.55807	1.359495	0.06601278	0.9882637
12	12	35.81602	0.2518339	28.56182	1.342309	0.06634668	1.0029103
13	13	35.67887	0.2567453	28.48257	1.331617	0.06959179	0.9780139
14	14	35.62027	0.2586596	28.41853	1.357934	0.07302252	1.0286034
15	15	35.56589	0.2605286	28.39269	1.363284	0.07153313	1.0447086
16	16	35.55408	0.2606200	28.36916	1.336615	0.07063410	1.0067380

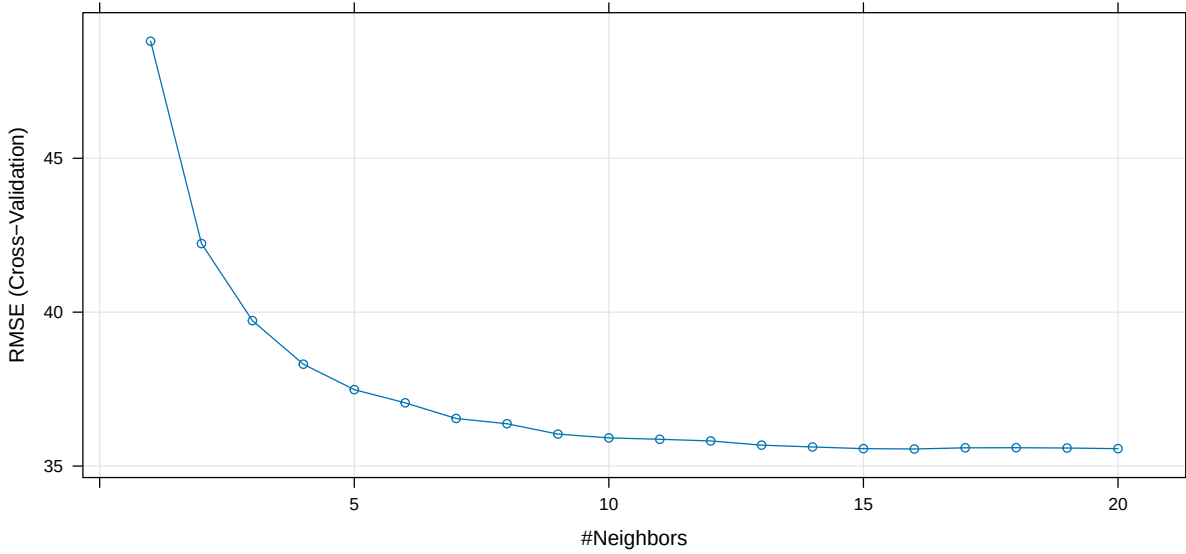


Figure 3: Cross-validation RMSE for KNN regression across K values

```

17 17 35.59253 0.2591857 28.38074 1.314678 0.07044019 0.9983158
18 18 35.59568 0.2585619 28.41177 1.259712 0.06891470 0.9778351
19 19 35.58604 0.2592335 28.36532 1.314577 0.07104216 1.0197781
20 20 35.56471 0.2599857 28.35991 1.333215 0.07200125 1.0473625

```

The best model occurs at $K = 16$. After this point, RMSE increases slightly, so I use $K = 16$ for the final KNN regression model.

At $K = 16$, the RMSE is approximately 35.5, meaning the typical prediction error of the model is about 35.5 minutes.

Comparison of Regression Techniques

I now compare K-Nearest Neighbors regression and Multiple Linear Regression.

Table 1: Training and Test MSE for Multiple Linear Regression and KNN Regression

Model	Training_MSE	Test_MSE
Multiple Linear Regression	1124.967	1292.095
KNN Regression ($K = 16$)	1071.956	1360.584

Based on the test errors, Multiple Linear Regression appears to perform slightly better than KNN regression. It also has better interpretability, which makes it more useful for explaining how each predictor relates to match duration.

Classification

The classification question is about predicting whether the match winner has high ace performance. I apply two classification techniques: Logistic Regression and Quadratic Discriminant Analysis (QDA). Using these

methods, I analyze how player and match variables such as winner height, court surface, match duration, winner rank, first-serve performance, and winner hand influence ace performance. To define the classification variable, I use the median number of winner aces as the cutoff. Matches above the median are classified as High Aces, while matches at or below the median are classified as Low Aces.

Boxplot of High vs. Low Aces

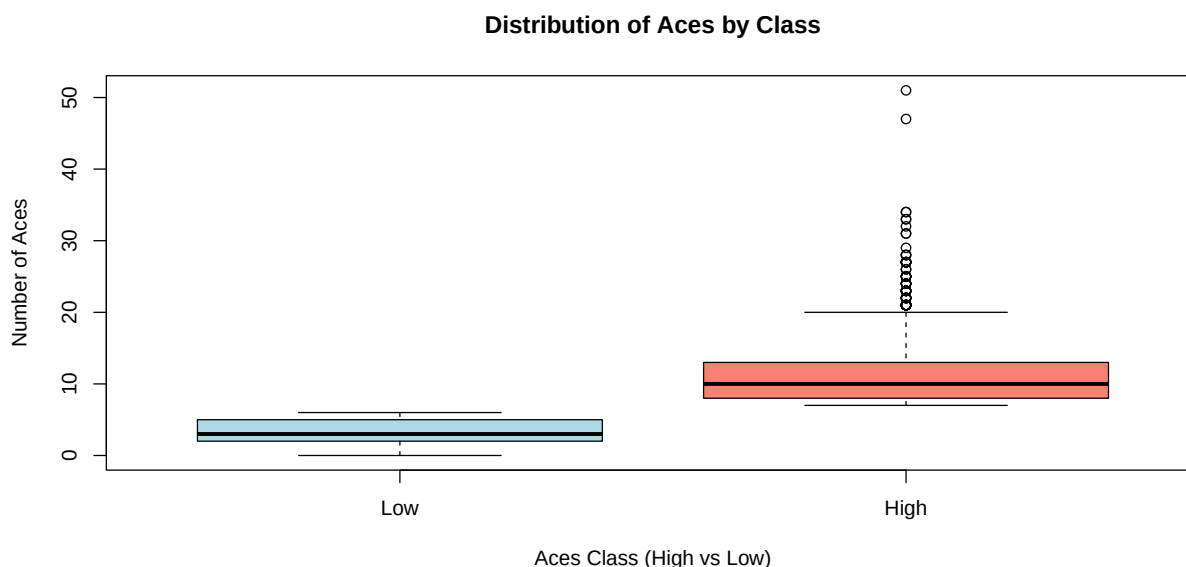


Figure 4: Distribution of Aces by Class

The boxplot shows a clear gap between the high and low ace classes, indicating a strong difference between the two groups. This suggests that the median split creates two distinct classes.

Boxplot of Aces by Surface

The boxplot suggests that grass courts tend to have the highest number of aces, followed by hard courts and then clay courts. This pattern is useful to keep in mind when interpreting the classification model results.

Scatterplot

This scatterplot compares ace performance with winner height, which is expected to be an important predictor of ace performance. The plot shows a positive relationship between winner height and number of aces, suggesting that taller players generally tend to serve more aces.

Summary of the response variable

Table 2: Summary of High and Low Ace Performance Classes

Category	Count
Low	1565

Category	Count
High	1451

The summary table shows that the classes are fairly evenly split between High and Low Aces, which is useful for classification modeling.

Logistic Regression

The first classification technique is logistic regression. This model estimates the probability that a match falls into the High Aces group using winner height, court surface, match duration, winner rank, first-serve performance, and winner hand as predictors. I also use the model to predict and plot how winner height and surface affect the probability of High Aces.

The logistic regression model is given by:

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \beta_6 X_6 + \beta_7 X_7$$

where $p_i = P(Y_i = 1)$ is the probability that the match falls into the High Aces category. Here, X_1 is winner height, X_2 and X_3 represent surface indicators, X_4 is match duration, X_5 is winner rank, X_6 is first-serve points won, and X_7 represents winner hand.

Call:

```
glm(formula = high_aces ~ winner_ht + surface + minutes + winner_rank +
     w_1stWon + winner_hand, family = "binomial", data = atp)
```

Coefficients:

```
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -2.004e+01  1.475e+00 -13.588  < 2e-16 ***
winner_ht    9.456e-02  7.751e-03  12.200  < 2e-16 ***
surfaceClay  -1.358e+00  1.099e-01 -12.355  < 2e-16 ***
surfaceGrass  1.952e-02  1.530e-01   0.128   0.8985
minutes      -1.174e-02  2.235e-03  -5.251  1.51e-07 ***
winner_rank  -6.553e-04  3.851e-04  -1.702   0.0888 .
w_1stWon      1.077e-01  8.094e-03  13.309  < 2e-16 ***
winner_handR -7.464e-03  1.335e-01  -0.056   0.9554
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

(Dispersion parameter for binomial family taken to be 1)

```
Null deviance: 3879.3  on 2799  degrees of freedom
Residual deviance: 2964.2  on 2792  degrees of freedom
(276 observations deleted due to missingness)
AIC: 2980.2
```

Number of Fisher Scoring iterations: 5

Results (Logistic Regression-Model Summary) The results indicate that several model variables are statistically significant predictors of ace performance. The model summary shows that winner height, clay surface, match duration, and first-serve performance have p-values below 0.05, suggesting that these factors are important in predicting whether a match falls into the High Aces category.

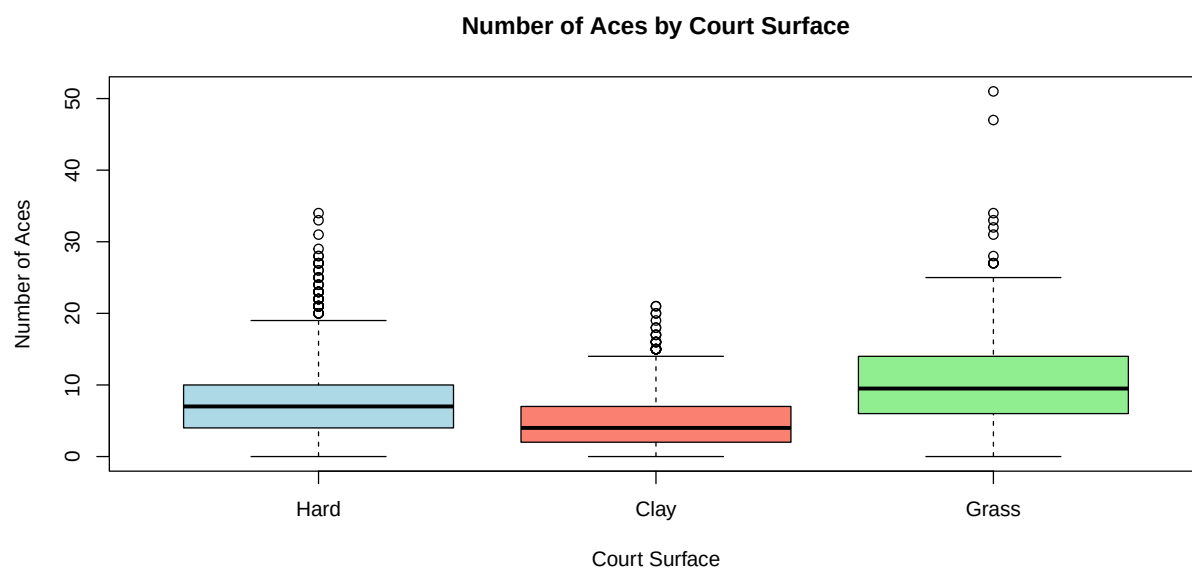


Figure 5: Number of Aces by Court Surface

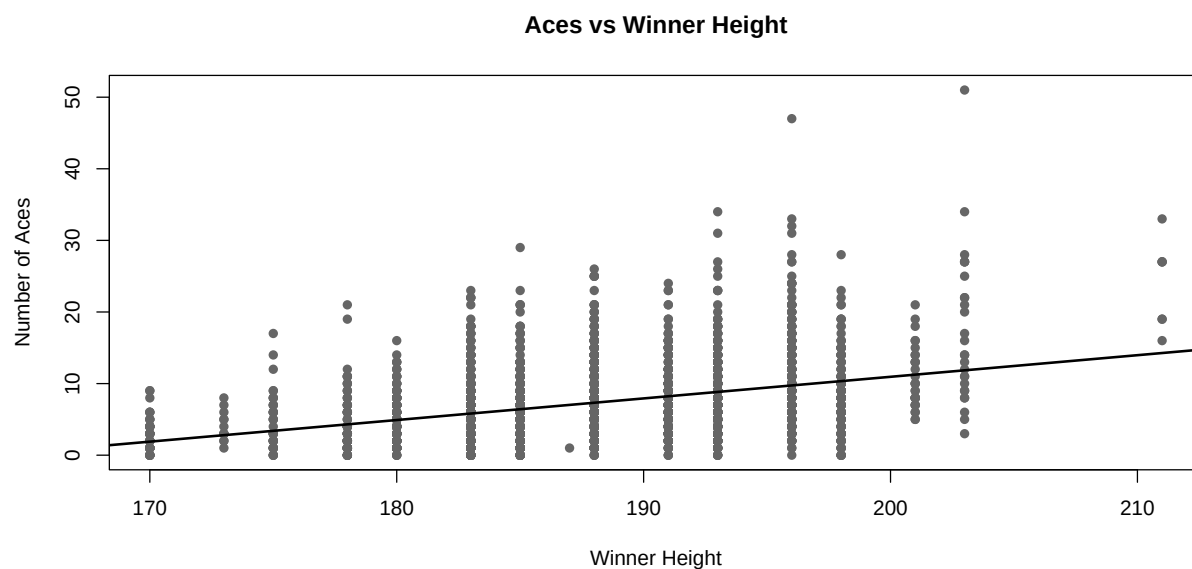


Figure 6: Relationship between winner height and number of aces

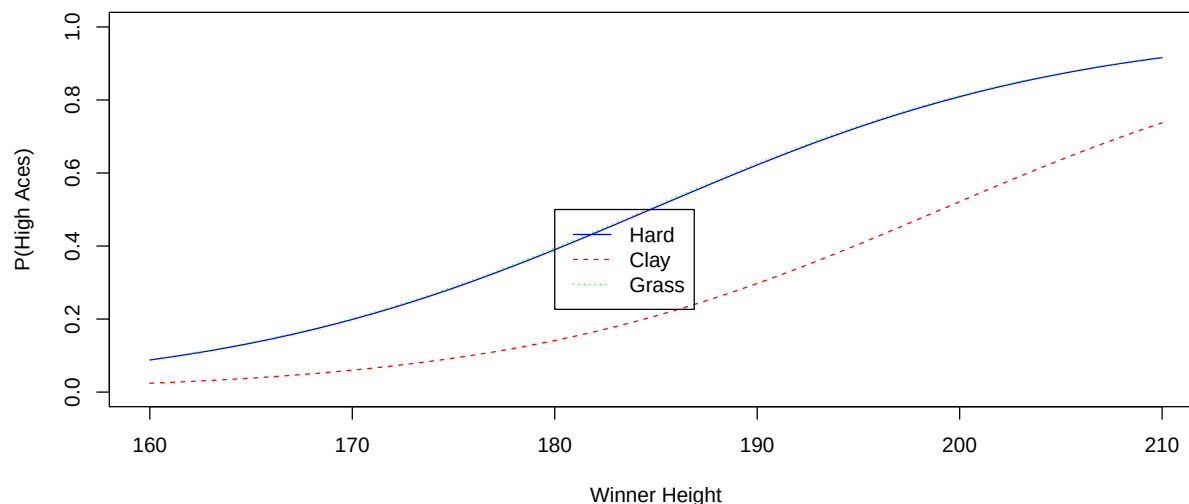


Figure 7: Predicted probability of high ace performance by winner height and surface

Results (Logistic Regression-Plot) I also tested the model by plotting the predicted probability of High Aces across winner height and surface. The plot shows that clay courts generally have a lower predicted probability of High Aces than hard and grass courts. Hard and grass courts appear relatively similar in the plot, which aligns with the model summary where the grass indicator is not statistically significant relative to hard courts. Winner height is also statistically significant, and the plot suggests that taller players generally have a higher predicted probability of being in the High Aces category.

Quadratic Discriminant Analysis (QDA)

This is the second classification technique. I consider the same match variables as the logistic regression model, but instead of only estimating probabilities, QDA directly predicts whether a match is classified as High or Low Aces. To carry out this analysis, I split the data into training and test sets, train the model on the training data, and evaluate how well it predicts unseen observations. This model may perform better than Logistic Regression if the data has nonlinear structure, because QDA is more flexible.

Call:

```
qda(high_aces ~ winner_ht + surface + minutes + winner_rank +
    w_1stWon, data = train)
```

Prior probabilities of groups:

```
      Low      High
0.5132653 0.4867347
```

Group means:

	winner_ht	surfaceClay	surfaceGrass	minutes	winner_rank	w_1stWon
Low	186.0457	0.3996024	0.06759443	106.7942	88.28529	32.58946
High	189.7883	0.1268344	0.15932914	125.4423	64.99686	42.24214

Results (QDA Model Summary) The QDA results show an almost even split between High and Low Aces. The model associates higher ace performance with taller players and grass surfaces, which is broadly

consistent with the logistic regression results showing that clay courts underperform relative to hard and grass courts. First-serve performance also appears positively related to High Aces, which is consistent with the logistic regression model. One difference is match duration: in the QDA results, longer matches are associated with higher ace performance, while the logistic regression coefficient for match duration is negative.

Table 3: Confusion Matrix for QDA Test Set Predictions

	Low	High	Sum
Low	329	105	434
High	146	260	406
Sum	475	365	840

[1] 0.7011905

The QDA model achieves 70.12% accuracy on the test set, indicating reasonably good classification performance.

Classification Comparison

I compare the classification models using cross-validation to evaluate which model performs better. Both models show similar trends, with taller players generally associated with higher ace performance. Both models also suggest that clay courts are associated with fewer aces.

Table 4: 10-Fold Cross-Validation Error and Accuracy for Logistic Regression and QDA

Model	CV_Error	CV_Accuracy
Logistic Regression	0.270	0.730
QDA	0.282	0.718

Results

Both models have similar cross-validation performance, with approximately 71–73% accuracy. QDA is usually more flexible and can fit nonlinear data better, but in this case Logistic Regression performs slightly better in cross-validation. This suggests that the simpler Logistic Regression model is sufficient for this classification task and performs slightly better than QDA.